Test 1.1 $G-3$

Ime in Priimek: ______________________________

Naloga 1

Zapiši s kotnimi funkcijami ostrih kotov v kotnih stopinjah in izračunaj:

a) $\dfrac{\cos 405^\circ + \sin(-150^\circ)}{\tan(-765^\circ)\cdot\cot(-240^\circ)}$

b) $\dfrac{\sin(1110^\circ - \cos 495^\circ}{\tan(-675^\circ)+\cot^2 300^\circ}$

c) $\dfrac{\cos(-300^\circ)\cdot\sin 480^\circ}{\tan 1140^\circ - \cot 135^\circ}$

d) $\dfrac{\cos 1080^\circ - \sin(-570^\circ)}{\frac{1}{2}\tan 225^\circ + \cot(-390^\circ)} + \sin^2(-30^\circ)$

e) $\dfrac{\sin^2 750^\circ + \cos 225^\circ}{\tan(-570^\circ)\cdot\cot 330^\circ}$

Naloga 2

Zapiši s kotnimi funkcijami ostrih kotov v radianih in izračunaj:

a) $$\frac{\sin\frac{17\pi}{4}+\cos\left(-\frac{5\pi}{3}\right)}{\tan\left(-\frac{9\pi}{2}\right)\cdot\cot^2\left(\frac{5\pi}{6}\right)}$$

b) $$\frac{\cos\frac{25\pi}{6}-\sin\left(-\frac{7\pi}{4}\right)}{\tan\frac{19\pi}{3}+\cot\frac{3\pi}{4}}\cdot\left(\tan\frac{4\pi}{3}-\cot\frac{5\pi}{4}\right)$$

c) $$\frac{\sin^2\frac{13\pi}{3}\cdot\cos\left(-\frac{11\pi}{6}\right)}{\tan\frac{15\pi}{4}-\cot\frac{4\pi}{3}}$$

d) $$\frac{\cos\left(-\frac{10\pi}{3}\right)+\sin\frac{23\pi}{4}}{\tan\frac{17\pi}{6}\cdot\cot\left(-\frac{5\pi}{4}\right)}+\sin\frac{13\pi}{6}$$

e) $$\frac{\sin\frac{29\pi}{6}-\cos\frac{19\pi}{4}}{\tan\left(-\frac{11\pi}{3}\right)+\cot^2\frac{7\pi}{4}}$$

Naloga 3

Zavrti točko $A(-\frac{1}{2},-\frac{\sqrt{3}}{2})$ za pravi kot okoli koordinatnega izhodišča v smeri urnega kazalca. Določi koordinate te točke.

Naloga 4

Na krožnici s polmerom 2 leži točka $T(\sqrt{3}, y<0)$.

a) Kakšni koordinati ima ta točka, če a) jo zavrtimo okoli koordinatnega izhodišča v nasprotni smeri gibanja urnega kazalca za kot $165°$?

b) če bi jo zavrteli v smeri gibanja urnega kazalca za kot $450°$?

Naloga 5

Poenostavi:

a) $(1+\tan^2 x)(1-\sin^2 x)$

b) $(1+\cot^2 x)(1-\cos^2 x)$

c) $(3\cos x+2\sin x)^2+(3\cos x-2\sin x)^2$

d) $\dfrac{1+\cos x}{\sin x}-\dfrac{\sin x}{1-\cos x}$

e) $\sin\left(x+\dfrac{\pi}{3}\right)+\sin\left(x-\dfrac{7\pi}{3}\right)$

f) $\cos\left(x+\dfrac{\pi}{4}\right)+\cos\left(x-\dfrac{9\pi}{4}\right)$

g) $\tan\left(x+\dfrac{\pi}{4}\right)\cdot\tan\left(x-\dfrac{5\pi}{4}\right)$

Naloga 6

Poenostavi:

a) $\dfrac{\sin 2x}{1+\cos 2x}$

b) $\dfrac{\cot^2 x-1}{2\cot x}\cdot\dfrac{\sin 2x}{2\cos^2 x-1}$

c) $\cot x-\tan x-2\cot 2x$

Naloga 7

a) Naj velja za oster kot x: $\tan x=\frac{12}{5}$. Izračunaj $\sin 2x$ in $\cos(x-\frac{\pi}{4})$.

b) Naj bo $0<x<\pi$ in $\cot x=\frac{3}{4}$. Izračunaj $\cos 2x$ in $\tan(x-\frac{\pi}{4})$.

c) Naj bo $\pi<x<2\pi$ in $\cos x=\frac{\sqrt{5}}{3}$. Izračunaj $\tan 2x$ in $\cot(x+\frac{\pi}{4})$.

d) Naj bo x topi kot in $\sin x=\frac{\sqrt{15}}{4}$. Izračunaj $\sin(x+\frac{\pi}{4})$ in $\cos(x-\frac{5\pi}{4})$.

e) Naj za ostra kota x in y velja $\sin x=\frac{3}{5}$ in $\cos y=\frac{5}{13}$. Izračunaj $\tan(x-y)$, $\sin\frac{x}{2}$, $\tan\frac{y}{2}$.

f) Naj bo $\frac{3\pi}{2}<x<2\pi$ in $\sin 2x=-\frac{24}{25}$. Izračunaj $\sin x$, $\sin\frac{x}{2}$.

g) Naj bo $\frac{\pi}{2}<x<\pi$ in $\cos 2x=\frac{40}{41}$. Izračunaj $\cos x$, $\cos\frac{x}{2}$.

h) Naj bo x oster kot in $\tan 2x=\frac{7}{24}$. Izračunaj $\tan x$, $\tan\frac{x}{2}$.

Naloga 8

Dokaži identiteto:

a) $\dfrac{\sin(3x)}{\sin x}-\dfrac{\cos(3x)}{\cos x}=2$

b) $\sin^4 x=\dfrac{3-4\cos(2x)+\cos(4x)}{8}$

c) $\tan(3x)=\dfrac{3\tan x-\tan^3 x}{1-3\tan^2 x}$

d) $\cos 4x=8\cos^4 x-8\cos^2 x+1$

Naloga 9

Dokaži enakost:

a) $\dfrac{\cos x \sin 2x + \cos 2x + 1}{\sin x + 1} \cdot (1 + \tan^2 x) = 2$

b) $\dfrac{\cos x + \cos 2x + 1}{\sin x + \sin 2x} = \cot x$

c) $\dfrac{\sin 2x}{\sin x}(1 + \tan^2 x) = \dfrac{2}{\cos x}$

Naloga 10

Zapiši kot produkt in izračunaj:

a) $\cos 105° - \cos 15°$

b) $\sin 105° + \sin 15°$

c) $\cos 15° - \sin 15°$

d) $\sin 5° + \sin 55° - \cos 25°$

e) $\sin 10° + \sin 50° - \cos 20°$

f) $\sin 8° + \cos 38° + \cos 158°$

Naloga 11

Zapiši kot vsoto ali razliko:

a) $\cos 7x \cos 5x$

b) $\sin 2x \sin 7x$

c) $\sin 6x \cos 4x$

d) $\sin(2x + 3y) \sin(2x - 3y)$

e) $\cos(x + y) \cos(x - y)$

f) $2 \sin(4x + y) \cos(4x - y)$

Naloga 12

Pokaži, da v trikotniku velja: $\sin 2\alpha + \sin 2\beta + \sin 2\gamma = 4 \sin \alpha \sin \beta \sin \gamma$

Naloga 13

Dokaži enakost:

a) $\dfrac{2 \sin 5x \cdot \sin 4x}{\cos x - \cos 9x} = 1$

b) $\dfrac{\cos 2x \cos 10x}{\cos 12x + \cos 8x} = \dfrac{1}{2}$

c) $\dfrac{\sin 4x + \sin 8x}{\cos 4x - \cos 8x} = \cot 2x$

Naloga 14

Nariši grafe funkcij:

a) $f(x) = 2\cos(x + \frac{\pi}{4}) - 1$

b) $f(x) = -2\sin(x + \pi) - 1$

c) $f(x) = \tan\left(\frac{x}{2} - \frac{\pi}{4}\right)$

a) $f(x) = |\cos(x - \frac{\pi}{3})| - 1$

b) $f(x) = -|\sin\left(\frac{x}{2} + \frac{\pi}{4}\right) + 2|$

c) $f(x) = |\cot\left(\frac{x}{3} - \frac{\pi}{6}\right)|$

Naloga 15

Reši enačbo:

a) $\cos(x + \frac{\pi}{2}) = \frac{\sqrt{2}}{2}$

b) $2\sin(x + \pi) = -\sqrt{3}$

c) $-3\tan(\frac{x}{3} + \frac{\pi}{6}) = \sqrt{3}$

d) $\cot(\frac{x}{2} + \frac{\pi}{3}) = \sqrt{3}$

e) $\cot^2 x - \frac{1}{9} = 0$

f) $\sin^2 x - \cos x = 1$

g) $\sin^2 x - \sin x = 2$

h) $\sin x = \sqrt{3}\cos x$

i) $\sin 2x \tan x = \frac{3}{2}$

j) $\cos 3x \sin 2x - \cos 3x = 0$

k) $\sin^2 x + 2\sin x\cos x - 3\cos^2 x = 1$

l) $\sin^2 x - 6\sin x\cos x - \cos^2 x = 3$

m) $cos2x = 5\sin^2 x - cos^2 x$

n) $\cos^3 x - \cos x = \sin^2 x$

1. Kote prevedemo v osnovne: - $405° = 360° + 45° \Rightarrow \cos(405°) = \cos(45°) = \frac{\sqrt{2}}{2}$ - $\sin(-150°) = -\sin(150°) = -\sin(180° - 30°) = -\sin(30°) = -\frac{1}{2}$ - $765° = 2 \cdot 360° + 45° \Rightarrow \tan(765°) = \tan(45°) = 1$ - $240° = 180° + 60° \Rightarrow \cot(240°) = \cot(60°) = \frac{\sqrt{3}}{3}$

   Števec: $\frac{\sqrt{2}}{2} + \left(-\frac{1}{2}\right) = \frac{\sqrt{2}-1}{2}$

   Imenovalec: $1 \cdot \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{3}$

   Celoten izraz:

$$\frac{\frac{\sqrt{2}-1}{2}}{\frac{\sqrt{3}}{3}} = \frac{\sqrt{2}-1}{2} \cdot \frac{3}{\sqrt{3}} = \frac{3(\sqrt{2}-1)}{2\sqrt{3}} = \frac{\sqrt{3}(\sqrt{2}-1)}{2}$$

$$\boxed{\frac{\sqrt{3}(\sqrt{2}-1)}{2}}$$

2. Kote prevedemo: - $1110° = 3 \cdot 360° + 30° \Rightarrow \sin(1110°) = \sin(30°) = \frac{1}{2}$ - $495° = 360° + 135° \Rightarrow \cos(495°) = \cos(135°) = -\frac{\sqrt{2}}{2}$ - $-675° = -2 \cdot 360° + 45° \Rightarrow \tan(-675°) = \tan(45°) = 1$ - $300° = 360° - 60° \Rightarrow \cot(300°) = -\cot(60°) = -\frac{\sqrt{3}}{3} \Rightarrow \cot^2(300°) = \frac{1}{3}$

   Števec: $\frac{1}{2} - \left(-\frac{\sqrt{2}}{2}\right) = \frac{1+\sqrt{2}}{2}$

   Imenovalec: $1 + \frac{1}{3} = \frac{4}{3}$

   Celoten izraz:

$$\frac{\frac{1+\sqrt{2}}{2}}{\frac{4}{3}} = \frac{1+\sqrt{2}}{2} \cdot \frac{3}{4} = \frac{3(1+\sqrt{2})}{8}$$

$$\boxed{\frac{3(1+\sqrt{2})}{8}}$$

3. Kote prevedemo: - $\cos(-300°) = \cos(300°) = \cos(360° - 60°) = \cos(60°) = \frac{1}{2}$ - $480° = 360° + 120° \Rightarrow \sin(480°) = \sin(120°) = \sin(60°) = \frac{\sqrt{3}}{2}$ - $1140° = 3 \cdot 360° + 60° \Rightarrow \tan(1140°) = \tan(60°) = \sqrt{3}$ - $135° = 180° - 45° \Rightarrow \cot(135°) = -\cot(45°) = -1$

   Števec: $\frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$

   Imenovalec: $\sqrt{3} - (-1) = \sqrt{3} + 1$

   Celoten izraz:

$$\frac{\frac{\sqrt{3}}{4}}{\sqrt{3}+1} = \frac{\sqrt{3}}{4(\sqrt{3}+1)} \cdot \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{3-\sqrt{3}}{4(3-1)} = \frac{3-\sqrt{3}}{8}$$

$$\boxed{\frac{3-\sqrt{3}}{8}}$$

4. Kote prevedemo: - $750° = 2 \cdot 360° + 30° \Rightarrow \sin(750°) = \sin(30°) = \frac{1}{2} \Rightarrow \sin^2(750°) = \frac{1}{4}$ - $225° = 180° + 45° \Rightarrow \cos(225°) = -\cos(45°) = -\frac{\sqrt{2}}{2}$ - $-540° = -2 \cdot 360° + 180° \Rightarrow \tan(-540°) = \tan(180°) = 0$ - $330° = 360° - 30° \Rightarrow \cot(330°) = -\cot(30°) = -\sqrt{3}$

   Števec: $\frac{1}{4} + \left(-\frac{\sqrt{2}}{2}\right) = \frac{1-2\sqrt{2}}{4}$

Imenovalec: $0 \cdot (-\sqrt{3}) = 0$
Deljenje z nič je nedefinirano.

$$\boxed{\text{Nedefinirano}}$$

5. Kote prevedemo: - $1080° = 3 \cdot 360° + 0° \Rightarrow \cos(1080°) = \cos(0°) = 1$ - $\sin(-570°) = -\sin(570°) = -\sin(360° + 210°) = -\sin(210°) = -(-\sin(30°)) = \dfrac{1}{2}$ - $150° = 180° - 30° \Rightarrow \tan(150°) = -\tan(30°) = -\dfrac{\sqrt{3}}{3}$ - $420° = 360° + 60° \Rightarrow \cot(420°) = \cot(60°) = \dfrac{\sqrt{3}}{3} \Rightarrow \cot^2(420°) = \dfrac{1}{3}$
Števec: $1 - \dfrac{1}{2} = \dfrac{1}{2}$
Imenovalec: $-\dfrac{\sqrt{3}}{3} + \dfrac{1}{3} = \dfrac{1-\sqrt{3}}{3}$
Celoten izraz:

$$\frac{\frac{1}{2}}{\frac{1-\sqrt{3}}{3}} = \frac{1}{2} \cdot \frac{3}{1-\sqrt{3}} = \frac{3}{2(1-\sqrt{3})} \cdot \frac{1+\sqrt{3}}{1+\sqrt{3}} = \frac{3(1+\sqrt{3})}{2(1-3)} = \frac{3(1+\sqrt{3})}{-4} = -\frac{3(1+\sqrt{3})}{4}$$

$$\boxed{-\frac{3(1+\sqrt{3})}{4}}$$